Actuator Cooling Coil

Flow, Thermal and Structural Evaluation

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ME396 | Lab 5

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Abstract

I evaluated a two-turn copper cooling coil using water, 40% propylene glycol in water, and mineral oil at inlet Reynolds numbers of 10^3, 10^5, and 10^7. The study combined steady flow calculations, a prescribed-flux thermal model, and static structural analysis with both straight leads fixed. All nine cases failed the required factor of safety of 2. Water at $Re = 10^3$ produced a pressure drop of 96.11 Pa, a flow rate of 0.3745 L/min, and an outlet temperature of 294.38 K. Its maximum von Mises stress was 232.49 MPa, giving a factor of safety of 1.2044. A pressure-suppressed calculation showed that thermal restraint controlled this case. I recommend water at $Re = 10^3$ as the operating condition for further development, with revised support compliance and a new structural assessment before accepting the design. The imposed heat flux fixes the heat removal near 36 W, so the present thermal model does not establish a coolant-dependent cooling advantage.

1 Introduction

The proposed cooling system serves a robotic shoulder actuator whose motor reaches approximately 80 °C after two hours of continuous operation. The design objective is to remove heat through a copper coil while maintaining modest coolant temperatures, a practical flow and pressure requirement, and a yield-based factor of safety of at least 2 [1]. The motor is a reference body rather than a simulated solid. Consequently, this study evaluates the loaded coil; it does not predict the motor's transient cooldown or its final temperature in service.

2 Problem Statement

The supplied geometry consists of the copper tube and its enclosed coolant volume. The tube has a 9.525 mm outside diameter, 0.762 mm wall thickness, and nominal 8.0 mm bore. It forms a two-turn helix with a 100.0 mm centerline diameter and 11.77 mm pitch. Each end has a 20 mm tangent lead. Flow enters lead B and leaves lead A. The coil surrounds an 89 mm diameter, 50.25 mm long motor with the specified nominal 0.75 mm clearance [1].

Table 1: Constant coolant properties specified at 20 °C [1]. PG40 denotes 40% propylene glycol by mass in water.

Coolant	ρ (kg/m^3)	c_p (J/kg K)	k (W/m K)	μ (Pa s)
Water	998.2	4182	0.598	0.001002
PG40	1032	3707	0.400	0.00438
Mineral oil	849	1886	0.134	0.0319

The copper was assigned Ansys General Materials *Copper Alloy*: density 8300 kg/m^3, Young's modulus 110 GPa, Poisson's ratio 0.34, thermal expansion coefficient 1.8×10^{-5} K^{-1}, thermal conductivity 401 W/(m K), and specific heat 385 J/(kg K). The library tensile yield strength of 280 MPa was used for all safety-factor calculations; the listed ultimate strength was 430 MPa.

2.1 Model connections and boundary conditions

The calculations used Ansys Workbench 2025 R1 Student. Fluent retained only the coolant volume, while Steady-State Thermal and Static Structural retained only the copper. Figure 1 shows the implemented one-way workflow.

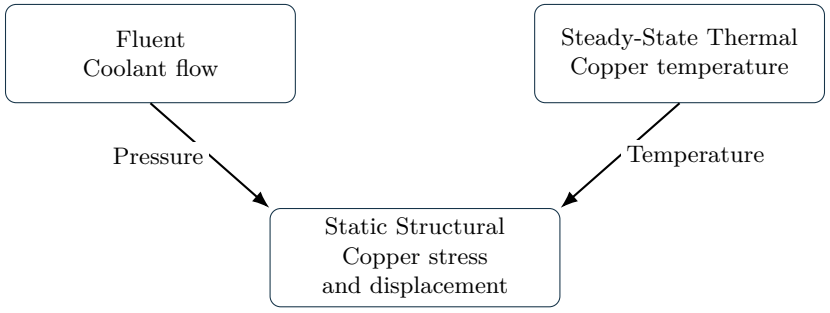


Figure 1: Implemented model connections. Both fluid and solid thermal models use prescribed heat fluxes with opposite signs. There is no imported-convection connection in this implementation.

With the instructor-approved direct-flux approach, I prescribed +2150 W/m^2 into the coolant in Fluent and −2150 W/m^2 on the copper's inner wall in Thermal. This replaces the assignment's imported-convection boundary condition. The negative solid-side flux is the corresponding heat loss in a separate domain, not an additional convection load. The same thermal solution applies to all nine cases because its material, geometry, and boundary conditions do not change.

Table 2: Applied boundary conditions and assumptions.

Model	Conditions
Fluent inlet	Uniform velocity from $Re\mu/(\rho d)$; temperature 293 K. Turbulent cases use 5% intensity and 0.008 m hydraulic diameter.
Fluent outlet	Pressure outlet, 0 Pa gauge; backflow temperature 293 K and the same turbulence specification.
Coolant wall	Stationary, no slip; heat flux +2150 W/m^2; zero wall thickness. Gravity and viscous heating are off.
Solid thermal	Outer curved tube surfaces, including straight leads, held at 80 °C; inner curved wall at −2150 W/m^2; annular end faces insulated.
Structural	Fixed support on both straight outer cylindrical lead faces. Imported inner-wall pressure from Fluent; imported copper temperature from Thermal. Stress-free reference temperature 25 °C. Linear-elastic static response.

The entered coolant temperature of 293 K is 19.85 °C; the imposed solid temperature of 80 °C is 353.15 K. These exact entered values were retained in the calculations rather than treating the assignment's rounded Celsius and kelvin values as identical. Structural pressure was mapped from `coolant_wall` to `copper_inner_wall` with scale 1 and offset 0. The reported pressure differences are gauge-pressure differences; no absolute-pressure margin is inferred from the zero-gauge outlet alone.

3 Method

3.1 Hand calculations and initial screening

Using the nominal bore $d = 0.008$ m,

$$v = \frac{Re\mu}{\rho d}, \qquad A = \frac{\pi d^2}{4} = 5.02655 \times 10^{-5}\ \mathrm{m}^2, \qquad \dot{V} = vA. \tag{1}$$

For water at $Re = 10^3$, $v = 1000(0.001002)/(998.2 \times 0.008) = 0.125476$ m/s and $\dot{V} = 0.378426$ L/min. Table 3 applies the same calculation to all nine specified cases. The Reynolds-number classification was used to choose the required laminar or turbulent model; curvature can still redistribute velocity within a laminar coil.

Table 3: Calculated inlet conditions and pre-simulation concerns. A dagger marks a case flagged for unusually high flow or speed.

Case	Fluid	Re	v (m/s)	$\dot{V}$ (L/min)	Model	Initial assessment
W3	Water	10^3	0.1254759	0.3784263	Laminar	Candidate
W5†	Water	10^5	12.54759	37.84263	Turbulent	High speed / flow
W7†	Water	10^7	1254.759	3784.263	Turbulent	Unphysical speed
G3	PG40	10^3	0.5305233	1.60002	Laminar	Candidate
G5†	PG40	10^5	53.05233	160.002	Turbulent	High speed / flow
G7†	PG40	10^7	5305.233	16000.2	Turbulent	Unphysical speed
O3†	Oil	10^3	4.696702	14.16492	Laminar	High flow demand
O5†	Oil	10^5	469.6702	1416.492	Turbulent	High speed / flow
O7†	Oil	10^7	46967.02	1.416492×10^5	Turbulent	Unphysical speed

Case IDs use W for water, G for PG40, and O for oil; the numeral gives the exponent of Re (for example, W3 is water at 10^3). Water and PG40 at 10^3 were plausible starting points. Even at 10^3, oil requires about 14.2 L/min because its high viscosity requires a much larger velocity to reach the same Reynolds number. Every 10^5 case requires substantially more flow, and the 10^7 velocities are outside a credible operating range for this liquid cooling loop. All nine were retained to complete the specified comparison.

3.2 Solver procedure

Fluent used a steady, pressure-based, double-precision solver with four processes and the energy equation enabled. The 10^3 cases used the laminar model. The 10^5 and 10^7 cases used realizable k–ε with standard wall functions. SIMPLE pressure–velocity coupling, least-squares cell-based gradients, second-order pressure, and second-order upwind transport discretization were used. Each material change included reassignment of the fluid properties and recalculation of inlet velocity.

The solutions used hybrid initialization and a maximum of 500 iterations. Residual targets were 10^{-5} for continuity, velocity and turbulence, and 10^{-8} for energy. The reported stopping iterations were 225 for 10^3, 210 for 10^5, and 195 for 10^7. Area-weighted inlet pressure and mass-weighted outlet temperature were monitored. Mass and energy flux reports provided additional checks; residual convergence alone was not treated as physical validation.

3.3 Evaluation metrics

The comparison uses $\Delta p = \bar{p}_{\mathrm{in}} - \bar{p}_{\mathrm{out}}$, measured volume flow $\dot{V} = \dot{m}/\rho$, outlet temperature, tube temperature range, thermal heat reaction, scoped and global von Mises stresses, and

$$P_h = \Delta p \dot{V}, \qquad \mathrm{FoS} = \frac{280\ \mathrm{MPa}}{\sigma_{\mathrm{VM,max}}}. \tag{2}$$

P_h is a hydraulic estimate, not electrical pump consumption. The safety requirement corresponds to a global stress limit of $280/2 = 140$ MPa.

3.4 Mesh and numerical checks

The coolant mesh used a 1.0 mm global size, 200 sweep divisions, and wall inflation with a 0.1 mm first layer, five maximum layers, and growth rate 1.2. It contained 49,000 elements and 51,858 nodes, below the course limit of 50,000 CFD cells. Maximum and mean skewness were 0.60073 and 0.14918; minimum and mean orthogonal quality were 0.62569 and 0.96904.

The copper used quadratic swept elements with a 1.5 mm global size. The initial 100-division mesh contained 3,900 elements and 25,683 nodes. Increasing the sweep to 120 divisions gave 4,680 elements and 30,783 nodes, below both solid-model limits of 32,000. This refined solid mesh was used for the nine-case comparison. The source mesh included triangles despite the all-quad request, producing mixed swept elements. Initial element quality ranged from 0.020763 to 0.26442 (mean 0.065226), so a favorable global-stress change cannot by itself establish complete mesh adequacy.

Table 4: Water, $Re = 10^3$: sensitivity to copper sweep divisions. Changes are relative to the 100-division mesh.

Quantity	100 divisions	120 divisions	Change (%)
Global stress (MPa)	233.43	232.49	−0.403
Support-surface stress (MPa)	205.05	205.12	+0.034
Outer-coil surface stress (MPa)	117.41	120.82	+2.904
Inner-coil surface stress (MPa)	126.21	130.90	+3.716
Maximum displacement (mm)	0.10396	0.10399	+0.029
Minimum FoS	1.1995	1.2044	+0.409

Table 4 supports limited axial-mesh stability of the global result and preserves the same pass/fail decision. It is not a full mesh-independence study: no independent CFD refinement, through-wall refinement, circumferential refinement, or stress-averaging study was completed.

Flow and energy consistency

The measured inlet mesh area was 4.975016×10^{-5} m^2, about 1.025% smaller than the nominal circular area. This accounts for the corresponding difference between calculated and reported inlet flow. For W3, $\dot{m} = 0.0062312144$ kg/s gives $\dot{V} = 0.374547$ L/min. Across the cases, normalized mass imbalance was at most 1.31×10^{-5}%. Energy imbalance was normalized by the wall heat input, rather than the much larger signed inlet and outlet energy terms; the values appear in Appendix A.

The Fluent heat input was 36.01355 W. For W3, the expected heat-only temperature rise is

$$\Delta T_b \approx \frac{\dot{Q}}{\dot{m} c_p} = \frac{36.01355}{0.0062312144(4182)} = 1.3820 \text{ K}, \tag{3}$$

consistent with the reported 1.38 K rise. This checks heat accounting within the imposed model, not the accuracy of omitted physical effects.

Near-wall limitation

At $Re = 10^5$, wall y^+ was approximately 15.69–41.01, with an area-weighted mean of 32.23. Part of the wall is below the commonly recommended standard-wall-function range of about 30–300 [2]. At 10^7, y^+ was approximately 1311–2568, with a mean of 2159. These ranges limit turbulent-result confidence. Viscous heating was off in every case, an increasingly important limitation as velocity and hydraulic demand rise [3].

4 Results

4.1 Nine-case comparison

Table 5: All nine cases using the refined solid mesh. All fail FoS ≥ 2. Heat removal and tube temperatures are common results of the imposed solid thermal boundary conditions.

	Flow				Thermal		Structural			
Case	v m/s	Δp kPa	$\dot{V}$ L/min	T_{out} K	$T_{\min}/T_{\max}$ °C	$\dot{Q}_{\text{th}}$ W	σ_{support} MPa	σ^{b}_{coil} MPa	σ_{global} MPa	FoS
W3	0.12548	0.09611	0.37455	294.38	79.996 / 80	36.144	205.12	130.9	232.49	1.2044
W5†	12.548	157	37.455	293.0138	79.996 / 80	36.144	205.14	130.97	232.53	1.2042
W7†	1254.8	7.1804×10^5	3745.5	293.00014	79.996 / 80	36.144	17681	24412	24412	0.01147
G3	0.53052	1.7763	1.5836	293.3567	79.996 / 80	36.144	205.12	130.9	232.49	1.2044
G5†	53.052	2901.7	158.36	293.0036	79.996 / 80	36.144	205.34	132.32	233.27	1.2003
G7†	5305.2	1.3271×10^7	15836	293[a]	79.996 / 80	36.144	3.2662×10^5	4.5131×10^5	4.5131×10^5	0.00062041
O3†	4.6967	114.53	14.02	293.0963	79.996 / 80	36.144	205.13	130.97	232.51	1.2042
O5†	469.67	1.8709×10^5	1402	293.001	79.996 / 80	36.144	4360.7	5891.7	5891.7	0.047525
O7†	46967	8.5567×10^8	1.4020×10^5	293[a]	79.996 / 80	36.144	2.1059×10^7	2.9100×10^7	2.9100×10^7	9.6220×10^{-6}

† Flagged in the hand calculations. Flow and pressure results confirm the high-demand concern for O3 and all 10^5 cases. All 10^7 cases also exceed the physical applicability of the simplified models. Their extreme numerical values are retained only to document screening failure, not as actual post-failure stresses or displacements.

[a] 293 K is the reported rounded outlet value for G7 and O7; it does not imply exactly zero heating. Heat-only temperature-rise estimates are 3.57×10^{-5} K and 9.63×10^{-6} K, respectively.

[b] The coil value is the larger maximum from the inner and outer curved-coil surface selections, excluding the straight lead surfaces. These selections reach the lead-to-coil junctions; they do not establish a stress value at a specified distance from the supports. They are therefore reported as *coil-surface* results rather than a verified remote-field stress. The global maximum governs FoS regardless of the lower scoped values.

Interpretation of the comparison

The maximum tube temperature remains 80 °C because it is prescribed. Similarly, the approximately 36 W thermal heat removal is imposed through the inner-wall flux. A lower coolant temperature rise at higher flow reflects greater heat-capacity flow rate, $\dot{m}c_p$, and does not show that more heat was removed. This model cannot rank coolants by independently predicted heat-transfer coefficient or motor cooling performance.

The low-Reynolds-number cases have nearly identical global stresses and safety factors. Increasing flow therefore does not remove the structural limitation caused by the rigid thermal restraint. In the most extreme cases, pressure loads produce stresses far beyond the copper yield strength; linear-elastic calculations no longer represent a physically realizable response. A converged numerical solution is not sufficient to accept those operating conditions.

4.2 Selected operating candidate: water at $Re = 10^3$

The following figures document the preferred operating candidate for further development, not a design that satisfies the safety requirement. Figure 2 shows pressure decreasing along the coil toward the zero-gauge outlet. Its local contour maximum of about 96.5 Pa is consistent with, but not identical to, the area-weighted inlet pressure of 96.11 Pa.

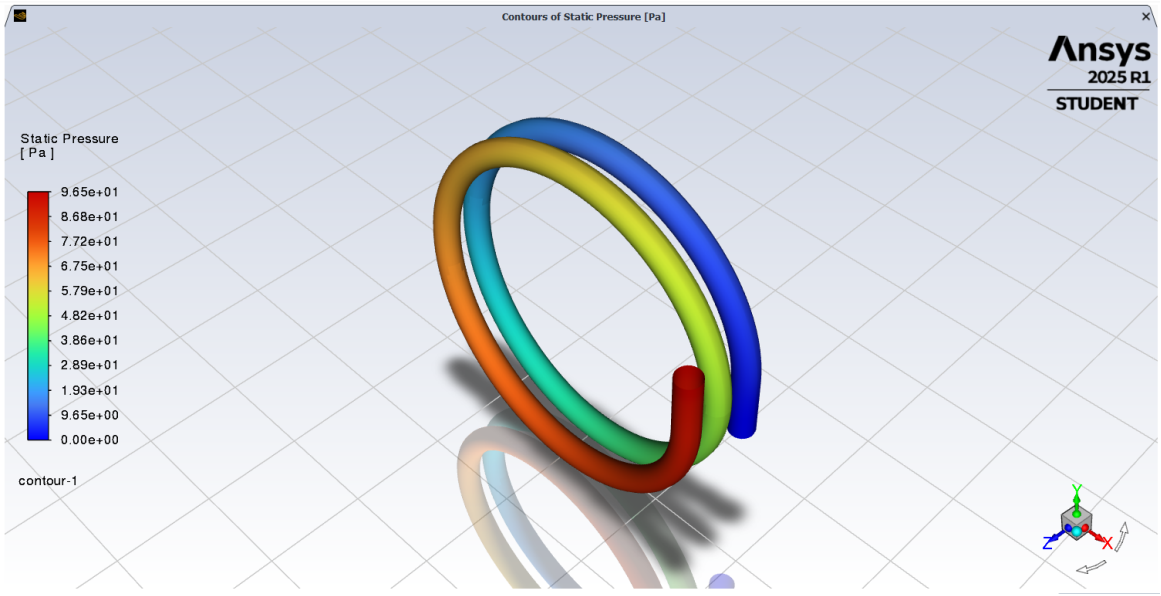


Figure 2: Water at $Re = 10^3$: static gauge-pressure contour along the coolant wall. The pressure-drop report gives 96.11 Pa.

Figure 3 shows the flow path through both turns. The paths curve across the tube as well as along it, consistent with curvature-induced redistribution. This visualization does not quantify a heat-transfer enhancement.

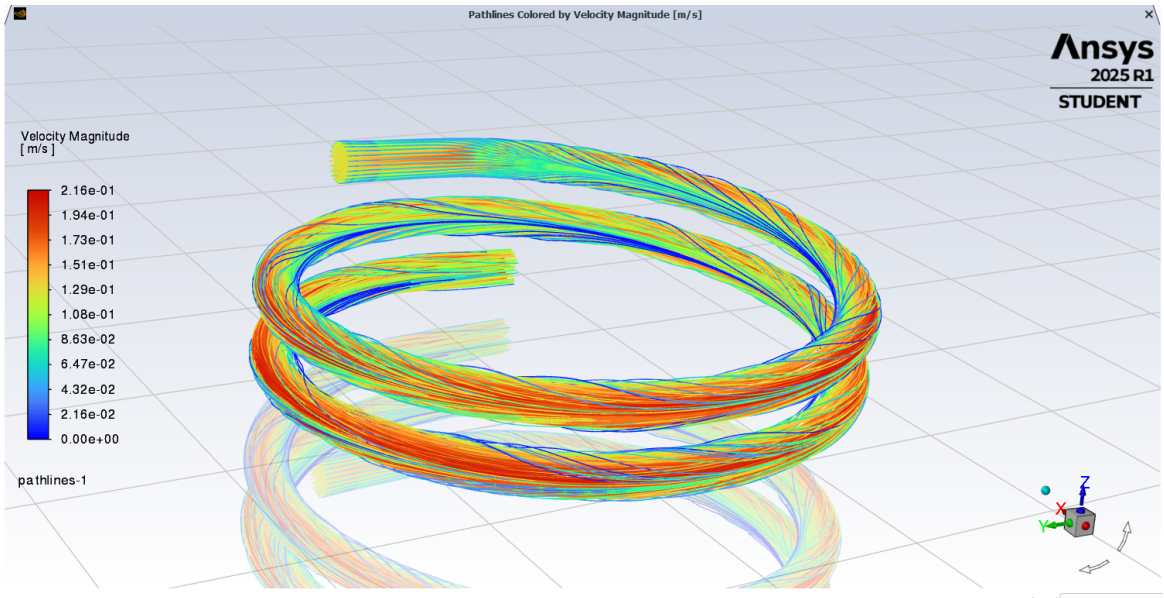


Figure 3: Water at $Re = 10^3$: pathlines colored by velocity magnitude, with a displayed range of 0–0.216 m/s. For the steady solution, pathlines represent streamlines. Faint images below the coil are display reflections.

4.3 Velocity distribution and tube temperature

The $y = 0$ axial plane intersects the tube at five locations, so separate circular sections in Figure 4 are expected. The warmer colors identify higher-speed fluid within each section, and the blue edges are consistent with the no-slip wall. The displaced velocity peaks show that the curved-pipe distribution is not uniformly axisymmetric.

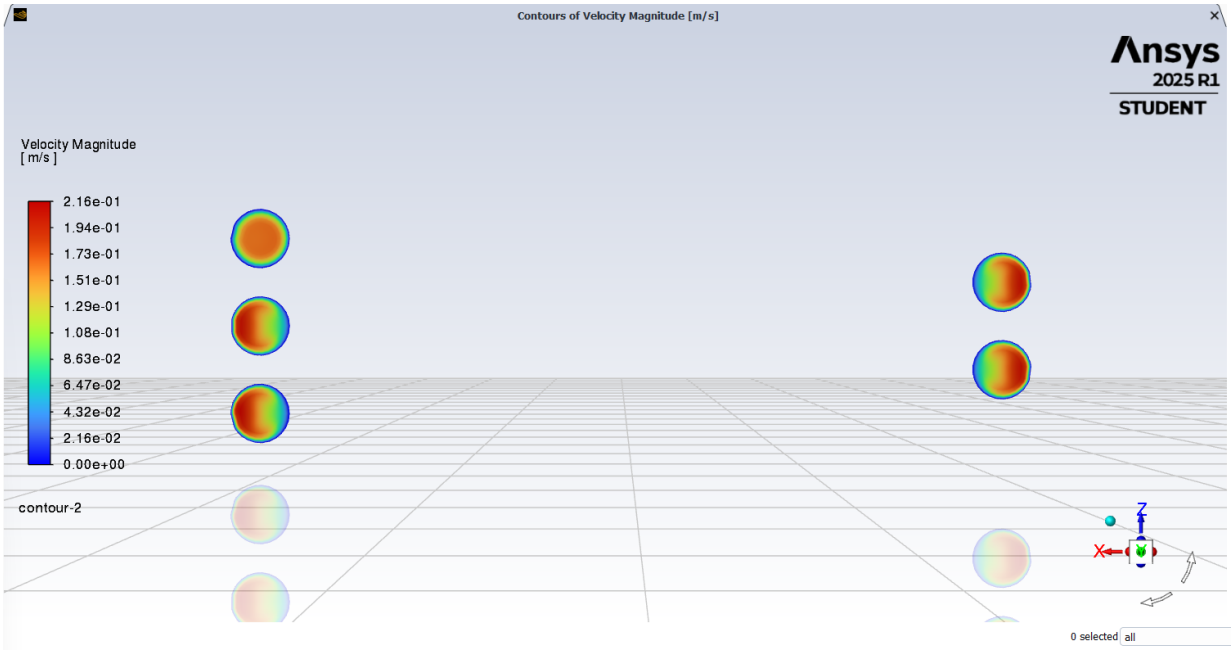


Figure 4: Water at $Re = 10^3$: velocity-magnitude contour on the $y = 0$ plane through the coil axis. The circles are separate intersections of one continuous tube with the plane.

The copper temperature is nearly uniform in Figure 5. The thermal solution gives a minimum of 79.996 °C and a maximum of 80.000 °C. A thin-wall conduction estimate, $\Delta T \approx q''t/k = 2150(0.000762)/401 = 0.00409$ K, agrees with this small range.

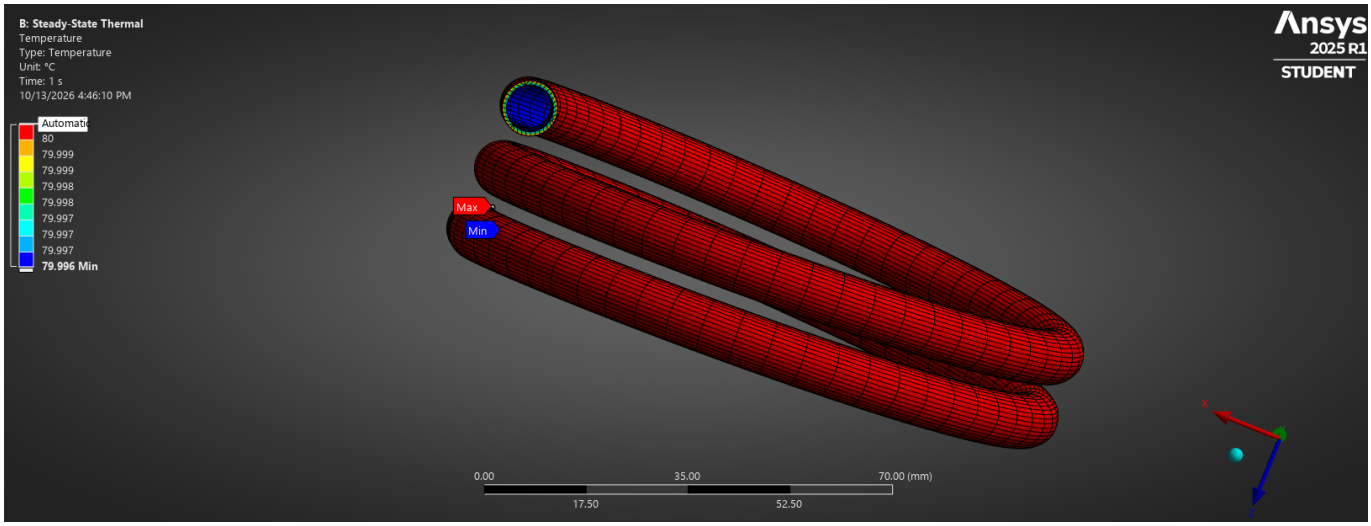


Figure 5: Copper temperature for W3 on the refined solid mesh: $T_{\min} = 79.996\,^{\circ}\text{C}$ and $T_{\max} = 80.000\,^{\circ}\text{C}$. The same imposed thermal solution is used in all cases.

The reaction at the prescribed-temperature boundary is 36.144 W, supplying the heat lost at the inner surface. Its 0.362% difference from the Fluent wall input is consistent with the different discretized wall areas: approximately 0.016811 m^2 for the solid and 0.016750 m^2 for the fluid. The reaction is not an independently predicted coolant cooling capacity.

4.4 Stress and safety factor

Figure 6 shows that the strongest stress concentrations occur in the lead regions under rigid restraint. The global maximum is 232.49 MPa, while the fixed-support surface maximum is 205.12 MPa. A surface-scoped maximum need not include the global critical location, so it cannot replace the global value in the acceptance check.

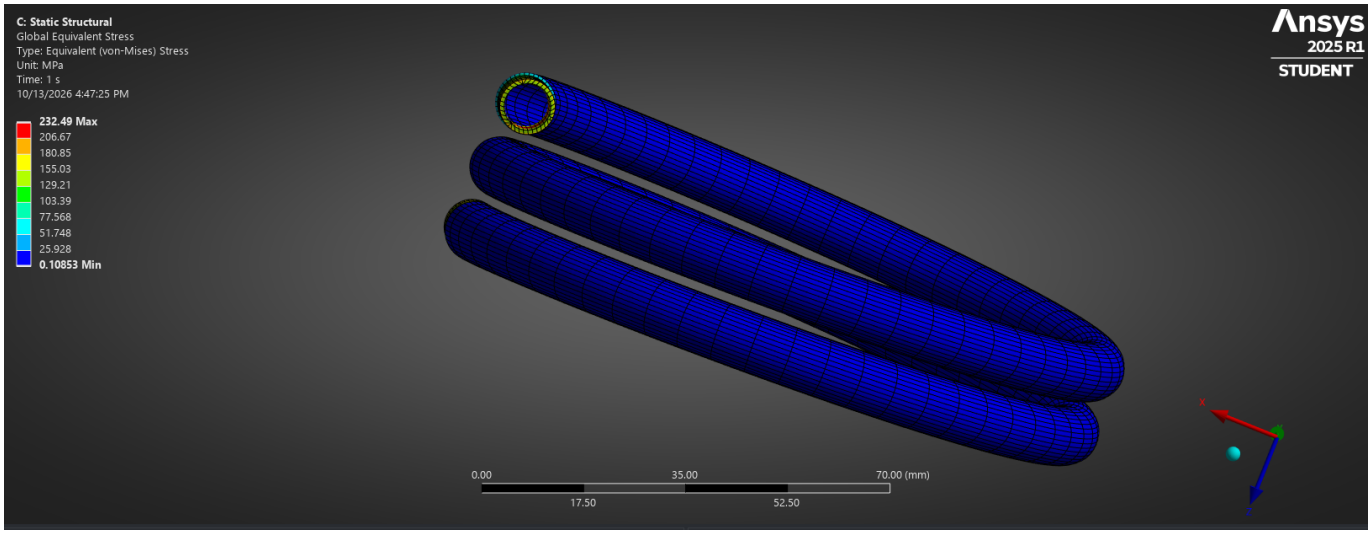


Figure 6: Water at $Re = 10^3$: global equivalent (von Mises) stress on the refined mesh. Maximum stress is 232.49 MPa.

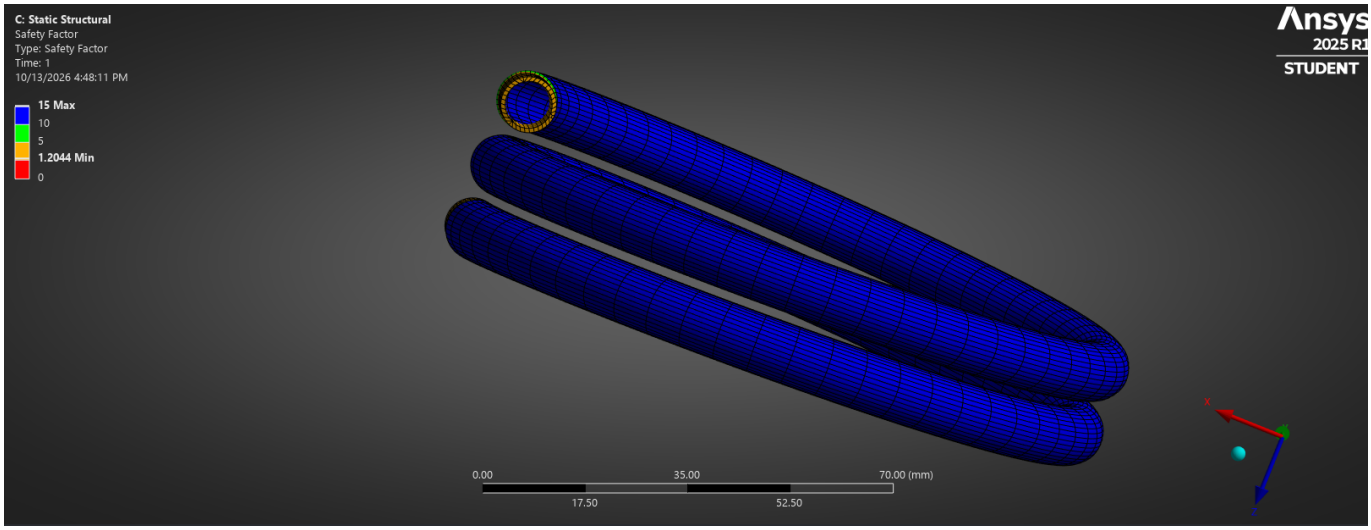


Figure 7: Water at $Re = 10^3$: minimum yield-based factor of safety is 1.2044. The displayed upper limit of 15 is a plotting limit; acceptance is controlled by the minimum.

The result in Figure 7 follows directly from $280/232.49 = 1.2044$. Although the computed stress is below the 280 MPa yield strength, it exceeds the 140 MPa allowable corresponding to FoS = 2. The design therefore fails the specified margin without necessarily predicting initial yielding in this low-flow case.

To separate the load contributions, I suppressed the imported pressure for W3. The maximum stress remained 232.49 MPa at the reported precision, while displacement changed from 0.10399 to 0.10397 mm. This supports thermal restraint as the controlling mechanism at low flow. A temperature rise of approximately 55 K from the stress-free condition corresponds to a free thermal strain $\alpha\Delta T \approx 9.9 \times 10^{-4}$. Fixing both leads prevents unrestricted expansion and concentrates stress near the constrained regions.

5 Discussion and Design Recommendation

5.1 Selection based on the stated requirements

No case qualifies as an acceptable final design because every minimum safety factor is below 2. Among the conditions worth retaining for redesign, W3 has the lowest calculated flow and pressure demand: 0.3745 L/min and 96.11 Pa for the modeled coil. Its coolant temperature rises only 1.38 K to 21.23 °C. These are modest bulk temperatures, but a liquid-phase margin still depends on the actual system absolute pressure and local wall conditions.

PG40 at $Re = 10^3$ requires 1.5836 L/min and 1.776 kPa, with a hydraulic estimate of 0.04688 W. Its use could be reconsidered if freeze protection becomes a system requirement. Oil at $Re = 10^3$ requires 14.0197 L/min and 114.53 kPa, with a hydraulic estimate of 26.76 W. Its higher flow demand provides no additional heat removal in this prescribed-flux model. No product-specific electrical insulation, compatibility, or freeze-temperature performance was evaluated.

For W3, $P_h = 0.000600$ W describes only the coil's calculated hydraulic load. It is not a prediction of milliwatt electrical pump consumption. Pump selection must account for the operating point, connecting hoses, fittings, reservoir and heat exchanger losses, efficiency, and a manufacturer's pump curve. The present results identify a low-demand starting point but do not select or validate a particular pump.

5.2 Why the extreme results are rejected

The 10^7 cases require velocities from about 1255 to 46,967 m/s and produce enormous pressure demands. These cannot be interpreted using the present constant-property, incompressible liquid model. O5 is also unsuitable, with a 187.09 MPa pressure drop and 97.976 mm computed displacement. Large negative local gauge pressures in several mapped loads further require absolute-pressure and cavitation checks. Neither cavitation nor pressure-dependent properties were modeled.

Viscous heating was disabled, and the turbulent wall-resolution limitations remain unresolved. Consequently, nearly unchanged outlet temperatures at high flow do not prove acceptable thermal operation. The very large elastic stresses and displacements in W7, G7, O5, and O7 indicate model and design failure, not the actual deformed shape or post-yield stress of a functioning coil. G5 also shows a substantial increase in displacement to 1.539 mm despite a nearly unchanged global stress maximum; similar peak stresses do not imply identical structural response.

5.3 Required design changes

I would first revise the support arrangement to permit controlled thermal expansion while maintaining sufficient restraint against rigid-body motion. A compliant connection or a guided/sliding lead is a candidate, followed by verification with realistic connection stiffness. Longer compliant lead transitions could also be evaluated. These are proposed changes; they were not simulated in this study.

The current W3 peak must decrease by approximately $1 - 140/232.49 = 39.8\%$ to meet FoS = 2 at the stated material strength. A revised model should evaluate support geometry, local mesh refinement, stress averaging, and cyclic thermal loading. It should also restore a coolant-dependent thermal boundary condition or use a conjugate heat-transfer model before claiming a heat-transfer benefit between coolants. The present model does not resolve the motor-to-coil gap or contact resistance.

6 Conclusion

The nine-case comparison rejects the current rigidly supported coil under the required safety-factor criterion. Water at $Re = 10^3$ is the preferred operating condition for further design work because it handles the specified approximately 36 W heat input with a small bulk temperature rise and the lowest flow and pressure demand. Its FoS of 1.2044 remains below 2, controlled primarily by restrained thermal expansion. A support redesign and renewed structural and thermal validation are necessary before recommending the coil for service.

A Supplementary Numerical Results

Table 6: Conservation checks and derived flow quantities. Imbalances use the solver-reported net fluxes, not sums of rounded component values.

Case	$\dot{m}$ kg/s	Mass imbalance % of inlet	Heat imbalance % of wall input	P_h W	ΔT_{heat} K
W3	0.006231214	1.30×10^{-5}	5.96×10^{-6}	0.00059996	1.382
W5	0.6231214	2.42×10^{-6}	0.000919	98.007	0.01382
W7	62.31208	2.89×10^{-6}	0.108	4.4823×10^{7}	0.0001382
G3	0.02723821	1.30×10^{-5}	0.000103	0.046883	0.3567
G5	2.723821	2.42×10^{-6}	0.00351	7658.6	0.003567
G7	272.3821	2.89×10^{-6}	0.417	3.5027×10^{9}	3.567×10^{-5}
O3	0.1983788	1.30×10^{-5}	0.000578	26.762	0.09626
O5	19.83788	2.42×10^{-6}	0.013	4.3716×10^{6}	0.0009626
O7	1983.788	2.89×10^{-6}	1.55	1.9994×10^{12}	9.626×10^{-6}

Here $\Delta T_{\text{heat}} = 36.01355/(\dot{m}c_p)$ is a derived heat-only estimate, not an additional simulated temperature. The largest wall-relative energy imbalance is 1.546% in O7, further limiting that extreme case.

Table 7: Additional structural outputs and mapped pressure extrema. Pressure extrema are local mapped wall values, not inlet or outlet averages.

Case	Outer coil stress (MPa)	Inner coil stress (MPa)	Maximum displacement (mm)	Mapped $p_{\min}$ MPa	Mapped $p_{\max}$ MPa
W3	120.82	130.9	0.10399	3.3621×10^{-7}	9.4455×10^{-5}
W5	120.87	130.97	0.14758	-0.0016633	0.1605
W7	24412	21450	366.55	-62.043	796.92
G3	120.82	130.9	0.10424	6.2138×10^{-6}	0.0017457
G5	126.6	132.32	1.539	-0.03074	2.9664
G7	4.5131×10^{5}	3.9655×10^{5}	6774.7	-1146.7	14729
O3	120.88	130.97	0.1325	0.00040065	0.11256
O5	5891.7	5123.8	97.976	-1.9821	191.26
O7	2.9100×10^{7}	2.5569×10^{7}	4.36810×10^{5}	-73934	9.4967×10^{5}

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[3] Ansys. *Fluent User's Guide*, Release 2025 R1, heat-transfer modeling and viscous heating. Available online. Accessed October 13, 2026.